

Urban Land-Use Efficiency

A Stable Metric for SDG 11.3.1 and Its Economic and Structural Drivers: 193 UN Member Countries x 1985-2020 Panel Data Analysis

Development Economics · SoSe 2026

Lecturer: Prof. Dr. Jürgen Bitzer

Ömer Furkan Çoban¹

Carl von Ossietzky University of Oldenburg, Applied Economics and Data Science, Oldenburg, Germany

Abstract

The official SDG 11.3.1 indicator, the ratio of land consumption rate to population growth rate (LCRPGR), is unstable: it divides by a growth rate that approaches zero and mixes an arithmetic numerator with a logarithmic denominator. We propose the Built-up per Capita Rate (BpCR), the log change in built-up area per person, which equals the difference of two log growth rates and is therefore well-defined for every country-period. Using a harmonised global panel of 193 UN member states (1985–2020) built from GHSL built-up and population grids under the Degree of Urbanisation, we first document where LCRPGR breaks down and show that, on identical data, replacing it with BpCR raises within-country explanatory power roughly nineteenfold. Estimating the drivers of BpCR with two-way fixed-effects and dynamic-panel GMM, we find that built-up area per capita rose in nearly two-thirds of country-periods, that per-capita land use is strongly path-dependent, and that within countries higher density, income, and net in-migration are each associated with less land consumed per resident, so economic growth coincides with densification rather than sprawl. The density association is strongest in the least-developed economies, while the income result holds across all development groups.

Keywords: SDG 11.3.1, Urban land-use efficiency, Built-up per capita rate (BpCR), LCRPGR, Dynamic panel GMM, GHSL / Degree of Urbanisation

1. Introduction

The world is urbanising rapidly: by the middle of this century close to two-thirds of humanity will live in cities, and almost all of the projected growth in built-up land will occur in developing countries (Gao and O'Neill, 2020). The scale of expansion is striking: every year the world's built-up area grows by roughly the size of the Netherlands (Seto et al., 2012). For development policy the central question is not whether

E-Mail Address: oemer.furkan.coban@uni-oldenburg.de (Ömer Furkan Çoban)

¹The author used the Claude Opus 4.8 and Gemini 3.5 large language models for improving the writing and for coding assistance; all analysis, results, and conclusions are the author's own.

cities expand, but how efficiently they do so, that is, how much additional land each new urban resident consumes. When built-up area grows in step with population cities stay compact; when it grows much faster they sprawl, paving over farmland for good, entrenching car dependence, raising the per-resident cost of infrastructure spread over more ground, and lifting per-capita emissions. Measuring land-use efficiency well is therefore a precondition for monitoring sustainable urbanisation.

The United Nations monitors this through target 11.3 of its Sustainable Development Goals (SDGs), whose official indicator 11.3.1 is the ratio of the Land Consumption Rate to the Population Growth Rate (LCRPGR) (UN-Habitat, 2025). Because LCRPGR is the agreed global yardstick, its statistical properties are not a technical detail: they determine how every country’s progress is judged and compared. Yet the indicator is fragile by construction: because the population-growth rate sits in the denominator, the ratio diverges and flips sign whenever population is roughly stable, so in exactly the countries where land-use efficiency matters most for policy it becomes uninformative.

This paper makes three contributions. First, it documents, on a harmonised global dataset, that LCRPGR is statistically unstable and that its instability is concentrated where population growth is near zero. Second, it proposes a simple, log-based alternative, the Built-up per Capita Rate (BpCR), the logarithmic growth rate of built-up area per capita. As the difference of two logarithmic growth rates it is finite for every country-period and directly interpretable, and it reproduces the policy signal of LCRPGR where the latter is well-behaved. Third, using the stable metric, the paper estimates the economic and structural drivers of per-capita urban land use on a panel of 193 United Nations member states observed at five-year intervals between 1985 and 2020.

The empirical analysis builds the panel from satellite-based settlement data under the global Degree of Urbanisation standard, so that built-up area and population are measured consistently. To address the well-known downward bias of the lagged dependent variable in dynamic fixed-effects models, the persistence of per-capita land use is estimated with Arellano-Bond and Blundell-Bond generalised method of moments (GMM) estimators. Three results stand out. Sprawl is the global norm rather than the exception: built-up area per capita rises in a clear majority of country-periods. Per-capita land use is strongly path-dependent, with about three-fifths of one period’s value carried into the next. And within countries, denser cities, net in-migration, and higher income are all associated with less land consumed per resident, indicating that, over time, economic growth is accompanied by densification rather than sprawl. These patterns largely hold across development groups, with the compact-city effect strongest in the least-developed economies.

2. Measurement Framework

This section sets out the conceptual core of the paper. We first define the official SDG 11.3.1 indicator and show why it is statistically unstable, and we then introduce the stable, log-based alternative used in the empirical analysis.

2.1. The SDG 11.3.1 Indicator and Its Limitations

Following the UN-Habitat metadata (UN-Habitat, 2025), the official indicator divides an arithmetic land consumption rate by a logarithmic population growth rate:

$$\text{LCRPGR}_i = \frac{\text{LCR}_i}{\text{PGR}_i}, \quad \text{LCR}_i = \frac{V_{i,t} - V_{i,t-z}}{V_{i,t-z}} \cdot \frac{1}{z}, \quad \text{PGR}_i = \frac{\ln(P_{i,t}/P_{i,t-z})}{z},$$

where, for country i , $V_{i,t}$ is the urban built-up area at the end of the period and $V_{i,t-z}$ the built-up area at the start; $P_{i,t}$ and $P_{i,t-z}$ are the corresponding urban populations; z is the length of the period in years; $\ln(\cdot)$ denotes the natural logarithm. The Land Consumption Rate (LCR) is thus the arithmetic growth of built-up area, expressed per year, and the Population Growth Rate (PGR) is the logarithmic (continuous) growth of population per year; the indicator LCRPGR is their ratio. This arithmetic-LCR / logarithmic-PGR pairing is not our reading but the metadata’s own operational specification: the UN-Habitat metadata and the accompanying JRC Land-Use-Efficiency tool define LCR as the simple annual rate of change of built-up area and PGR as the continuous (exponential) population-growth rate (UN-Habitat, 2025; Corbane et al., 2017), and Nicolau et al. (2019) show formally that mixing the two scales is what destabilises the ratio. Two structural problems follow. **(1) Division by a near-zero denominator.** When population is roughly stable ($\text{PGR} \approx 0$) the ratio diverges towards $\pm\infty$ and flips sign, so small, ordinary demographic differences produce wild, uninterpretable values. **(2) Arithmetic-versus-logarithmic asymmetry.** The numerator is

an arithmetic growth rate while the denominator is logarithmic, so the ratio is not symmetric and does not aggregate cleanly over time (Nicolau et al., 2019). The result is a metric whose tails are dominated by an arithmetic artefact rather than by real land-use behaviour.

2.2. A Stable Metric: The Built-up per Capita Rate (BpCR)

The UN-Habitat metadata itself concedes that the ratio is hard to interpret and offers built-up area per capita (V/P) as a secondary indicator (UN-Habitat, 2025). That quantity is a level, a snapshot at one point in time. Following Lu and Weng (2026), we turn it into a trend by taking its annual growth in logarithmic form:

$$\text{BpCR}_{i,t} = \frac{1}{z} \ln \left(\frac{V_{i,t}/P_{i,t}}{V_{i,t-z}/P_{i,t-z}} \right) = \text{LCR}_{i,\log} - \text{PGR}_{i,\log},$$

where $\text{LCR}_{i,\log} = \frac{1}{z} \ln(V_{i,t}/V_{i,t-z})$ and $\text{PGR}_{i,\log} = \frac{1}{z} \ln(P_{i,t}/P_{i,t-z})$ are the logarithmic growth rates of built-up area and population. Here $\text{PGR}_{i,\log}$ is identical to the PGR of the official indicator above, which is already logarithmic, whereas $\text{LCR}_{i,\log}$ is the logarithmic counterpart of the arithmetic LCR_i in that indicator. Replacing the arithmetic LCR with its logarithmic form is precisely what removes the arithmetic-versus-logarithmic asymmetry that destabilises LCRPGR.

Because BpCR is the **difference of two logarithmic growth rates**, it is (i) finite and well-defined for every country-period (no division by ~ 0), (ii) symmetric (densification and sprawl are mirror images around zero), (iii) additively decomposable, and (iv) directly interpretable: $\text{BpCR} > 0$ means sprawl (more land per resident), $\text{BpCR} < 0$ densification, and $\text{BpCR} = 0$ land growing exactly in step with population. BpCR reproduces the policy signal of LCRPGR where the latter is well-behaved, but without the instability; it complements rather than contradicts the official indicator.

In the empirical sections we benchmark BpCR against the logarithmic form of the ratio, $\text{LCRPGR}_{\log} = \text{LCR}_{i,\log}/\text{PGR}_{i,\log}$ (*LCRPGR (log)*). This is a deliberately conservative benchmark: already free of the arithmetic-versus-logarithmic asymmetry, it differs from BpCR only in being a ratio rather than a difference, so any instability it shows comes purely from dividing by a near-zero growth rate; the official arithmetic ratio would be more unstable still.

3. Literature Review

The literature on urban land use splits into two strands. One asks what *drives* urban expansion; the other scrutinises how land-use efficiency is *measured*. This paper sits at their intersection.

3.1. Drivers of Urban Land Expansion

Seto et al. (2011) synthesise 326 remote-sensing studies of global urban land expansion between 1970 and 2000. They document that urban land has grown faster than urban population almost everywhere, and that the expansion is driven mainly by income in most regions and by population growth in India and Africa. Oueslati et al. (2015) move from description to econometrics, estimating the determinants of sprawl on a panel of 282 European cities (1990, 2000, 2006) within a monocentric-city framework. They find that rising income and population both enlarge the urban footprint, and that cities keep spreading even where population is flat. Together these studies establish income and population as the leading drivers of urban land use, but they work mostly across cities or countries and at a single cross-section, leaving open whether the same relationships hold within a country as it develops over time.

3.2. Critiques of the Indicator: Formula and Data

Nicolau et al. (2019) assess the SDG 11.3.1 formula itself, comparing the official LCRPGR with a land-use-efficiency-derived alternative at the municipality level in mainland Portugal. They conclude that the LCRPGR formula is structurally unstable and that a logarithmic, efficiency-based alternative is more robust. Lu and Weng (2026) broaden the question to data sensitivity, evaluating land-use efficiency across six built-up and four population datasets for 176 countries over 2000–2020. They show that efficiency estimates swing widely with the choice of dataset and argue that a log-based framework is needed for reliable monitoring. In the same vein, Zhong et al. (2025) propose a more reliable, complete, and equitable global land-use-efficiency assessment, again stressing that data and methodological choices shape the resulting picture of urban sustainability. This strand establishes that the choice of metric, and not only the choice of data, can change the conclusions drawn about sustainable urbanisation.

3.3. Research Gap and Contribution

Our contribution sits at the intersection of the two strands. On measurement, we adopt the log-based metric that both [Nicolau et al. \(2019\)](#) and [Lu and Weng \(2026\)](#) recommend, in the form of BpCR. On method, we make it dynamic: a two-way fixed-effects panel with Arellano-Bond and Blundell-Bond GMM, estimated on 193 United Nations member states over 1985–2020, that measures the persistence of per-capita land use and its economic drivers.

4. Data

4.1. Data Sources

The empirical analysis draws on three official, global data providers, the same sources used in international SDG monitoring. Built-up area, population, and the urban mask come from the **Global Human Settlement Layer (GHSL)** of the European Commission’s Joint Research Centre ([Pesaresi et al., 2024](#); [Schiavina et al., 2023](#)). Within GHSL we use three products: GHS-BUILT-S for built-up surface, classified from Landsat and Sentinel-2 imagery by machine learning; GHS-POP, which disaggregates census counts onto that built-up surface; and GHS-SMOD, which applies the Degree of Urbanisation (DEGURBA) method to flag urban cells. The layers are available at five-year epochs; we use the observed epochs from 1980 to 2020 (the 1980 layers serve only as the base period for the 1985 growth rates, so the estimation sample itself spans 1985–2020) and discard the 2025 and 2030 modelled projections. Administrative units come from the **Global Administrative Unit Layers (GAUL)** of the UN Food and Agriculture Organization ([FAO, 2024](#)), which provide UN-validated, politically neutral country boundaries with global coverage. Socio-economic controls come from the **United Nations**: GDP per capita in constant 2020 US dollars from UN SNAAMA and international net migration as a share of population from UN DESA World Population Prospects 2024 ([UN SNAAMA, 2025](#); [UN DESA WPP, 2024](#)). For the heterogeneity analysis we classify countries into United Nations development groups, developed economies, developing economies, and least-developed countries (LDCs), using the UN M49 regional classification together with the UN Committee for Development Policy LDC list ([UN M49 / UN-CDP, 2024](#)) (obtained through the `countrycode` R package).

Table 1 summarises the technical characteristics of each dataset.

Table 1: Dataset specifications. GHSL grids use the World Mollweide projection (ESRI:54009); built-up and population at 100 m, the SMOD urban mask at 1 km. Vector and tabular sources show their native form.

Dataset	Provider	Type	Resolution	CRS	Years	Variable
GHS-BUILT-S R2023A	EC JRC	Raster	100 m	Mollweide (54009)	1975-2030, 5-yr	Built-up area (m ² /cell)
GHS-POP R2023A	EC JRC	Raster	100 m	Mollweide (54009)	1975-2030, 5-yr	Population (persons/cell)
GHS-SMOD R2023A	EC JRC	Raster	1 km	Mollweide (54009)	1975-2030, 5-yr	Degree of Urbanisation (codes 10-30)
GAUL 2024/2025	FAO	Vector	Levels 0-2	WGS84 (EPSG:4326)	2024/2025	Admin boundaries
UN SNAAMA	UN Stats	Tabular	Country-year	–	1970-2024	GDP p.c. (const. 2020 US\$)
UN DESA WPP 2024	UN DESA	Tabular	Country-year	–	1950-2023	Net migration (% of pop.)
UN M49 / CDP	UN	Classification	Country	–	2024	Development group

4.2. Constructing the Panel Following the UN Metadata

A central design choice is that we build the panel exactly as the SDG 11.3.1 metadata prescribes ([UN-Habitat, 2025](#)), so that our BpCR and the official LCRPGR are computed on identical inputs and the comparison is like-for-like. The construction follows the metadata stages.

Figure 1 illustrates the construction for one example window, with each layer drawn over a satellite basemap. Panel (a) is the basemap for reference; panel (b) shows the GHS-BUILT-S surface as the built-up share of each cell; panel (c) the GHS-SMOD Degree of Urbanisation, which classifies each cell from rural to urban centre; and panel (d) the urban-masked built-up that actually enters the analysis, retaining built-up only in cells classified as urban (SMOD at least 21). The same masking is applied to population, so that both components are measured on a consistent urban footprint.

Analysis area. SDG 11.3.1 is an urban (city-scale) indicator, measured inside cities rather than over whole national territories. We delineate the urban area with DEGURBA on the GHS-SMOD grid (a cell is urban when its settlement code is at least 21, that is, urban centres and dense or semi-dense towns and suburbs) and measure all built-up and population quantities only over these urban cells. The built-up and population layers are native 100 m grids and the SMOD urban mask is a 1 km grid; we resolve the resolution mismatch by aggregating the 100 m layers to the 1 km SMOD grid before masking, so that all three layers share a common cell.

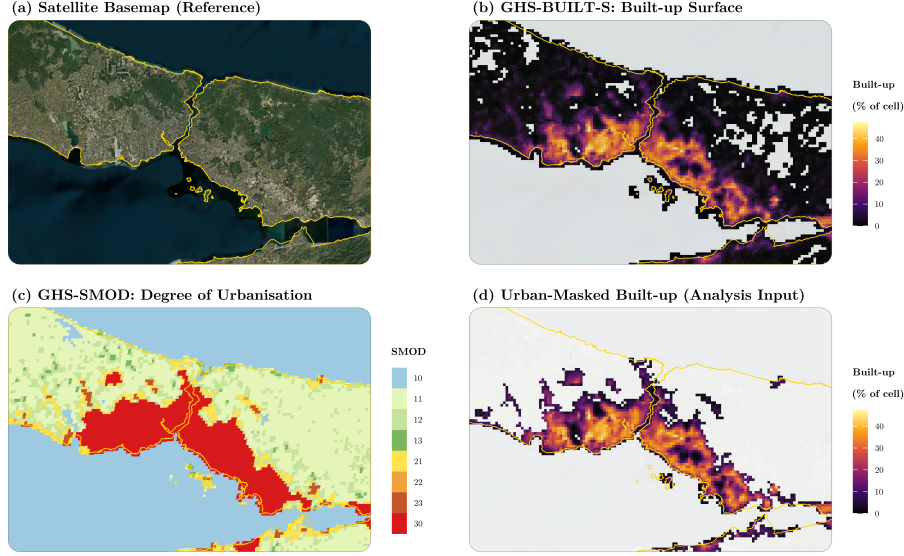


Figure 1: Building the analysis input following the SDG 11.3.1 metadata (Istanbul / Bosphorus window), each layer drawn over an Esri satellite basemap. (a) Basemap. (b) GHS-BUILT-S built-up surface (native 100 m, shown here as the built-up share aggregated to 1 km cells for display, to match the 1 km SMOD grid). (c) GHS-SMOD Degree of Urbanisation (1 km); cells coded 21 or higher count as urban. (d) Urban-masked built-up, the input actually used: panel (b) kept only where panel (c) is urban. The gold line is the GAUL country boundary.

Components and aggregation. Our unit of analysis is the country, so we aggregate the urban cells to the national level, aggregating the *components* rather than the metric: for each country-period we sum urban built-up area V and urban population P before forming any rate. This follows the metadata’s aggregate-then-rate rule; averaging city-level ratios would let a small town with a near-zero population denominator dominate the national figure, whereas summing components first keeps the national rate well-defined. Boundaries are held fixed at their current GAUL definition for every epoch, so each country is a constant spatial footprint and the within-country changes we study reflect land-use dynamics rather than shifting borders; both the GHSL grids and the UN series are keyed to present-day states.

Rates and indicators. From these national totals we form the Land Consumption Rate (arithmetic growth of built-up area), the Population Growth Rate (logarithmic growth of population), their ratio LCRPGR, the built-up-per-capita level, and BpCR as the logarithmic growth of built-up area per capita (Section 2).

Control variables. The two urban controls are computed from the same GHSL/SMOD layers. **Urban density** is national urban population divided by urban land area, the area of urban cells (distinct from, and about ten times larger than, the built-up surface in the dependent variable), used in logs. **Urban population share** is urban population divided by national total population. The remaining controls come from international sources: **GDP per capita** (constant 2020 US dollars, logs) from UN SNAAMA and **net migration** (international, share of population) from UN DESA WPP 2024. All variables align to the same country-period grid.

The resulting dataset is an unbalanced panel, complete for most but not all country-periods, of 193 United Nations member states across eight five-year epochs from 1985 to 2020, giving 1,538 country-periods with an observed BpCR. Because BpCR is a growth rate, each epoch’s value is computed against the preceding five-year grid: the 1985 BpCR uses the 1980 GHSL layers as its base, the 1990 value uses 1985, and so on, so the eight 1985–2020 BpCR observations are formed from the 1980–2020 grids. The common working sample used across specifications keeps country-periods with the lagged dependent variable and all four controls present (complete-case filtering), which reduces the sample to 1,499 country-period observations; this is the sample size reported for the static and fixed-effects columns in Section 7. The GMM estimators are fit on this same working sample, but, as detailed in Section 6, their internal instrument construction does not drop country-periods with a valid dependent variable, so the Arellano-Bond and Blundell-Bond observation counts differ from 1,499 and from each other for reasons unrelated to missing data.

5. Descriptive Statistics

Table 2: Summary statistics, 1985–2020 (country-periods with an observed BpCR). LCRPGR (log) is the log-form ratio benchmark; BpCR is the log growth of built-up area per capita.

Variable	N	Mean	SD	Min	Median	Max
BpCR	1538	0.004	0.017	-0.114	0.004	0.110
LCRPGR (log)	1538	1.304	17.532	-568	1.031	281
ln(Urban Density)	1538	7.581	0.572	5.719	7.606	9.618
Urban Pop. Share	1538	0.673	0.167	0.122	0.702	1.000
ln(GDP per capita)	1501	8.394	1.517	4.795	8.298	12.139
Net Migration (% Pop)	1538	-0.112	2.268	-70.990	-0.043	8.622

Table 2 reports summary statistics over the full 1985–2020 sample, and even before any regression the raw data reveal why the choice of metric matters. Over the 1,538 country-periods, LCRPGR (log) has a standard deviation of about 18 and ranges from roughly -568 to 281: its division by a near-zero population-growth rate produces enormous, sign-flipping outliers. BpCR, by contrast, stays in a tight band with a standard deviation of about 0.017, roughly 1,020 times narrower: the same underlying data, yet one metric is dominated by an arithmetic artefact and the other is not.

Sprawl is the global norm rather than the exception. BpCR is positive in 63% of country-periods, and 129 of 193 countries (67%) have a positive average BpCR over the period, meaning built-up area per capita is rising: cities are consuming more land per resident. The median country-period BpCR is +0.0041, and the official arithmetic LCRPGR ratio exceeds one (its sprawl threshold) in 56% of cases. The qualitative message is the same under either metric; the difference is that BpCR delivers it without the numerical instability.

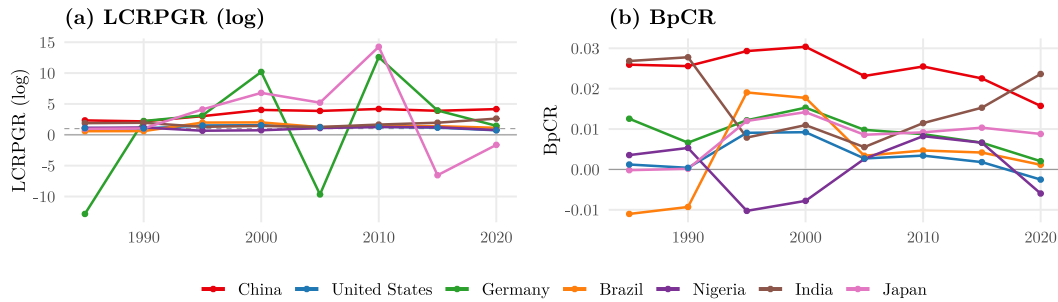


Figure 2: Country trajectories for seven regional representatives, 1985–2020. (a) LCRPGR (log) swings erratically across its zero and unity reference lines; (b) BpCR is smooth on the same data, with the fastest-urbanising economies showing the strongest early sprawl.

Country trajectories make the contrast concrete (Figure 2). For a set of regional representatives, LCRPGR (log) swings erratically from one period to the next, crossing zero and spiking to large positive and negative values that are hard to read as land-use behaviour. The BpCR series for the same countries are smooth and interpretable: most drift gradually towards balance, while the fastest-urbanising economies show the strongest sprawl earlier in the period before easing.

6. Empirical Strategy

Research question. We ask two linked questions. The substantive one is what drives *within-country* changes in per-capita urban land use over time, and how persistent they are. The measurement one is whether the official SDG 11.3.1 indicator captures this reliably, or whether the choice of metric changes the conclusions. We formalise these in five hypotheses (Table 3).

Table 3: Hypotheses

	Hypothesis	Expected
H1	BpCR captures land-use efficiency more reliably than LCRPGR	Better Fit
H2	Higher urban density is associated with less land per capita	—
H3	Higher income is associated with less land per capita	—
H4	International net in-migration densifies cities	—
H5	Per-capita land use is path-dependent	+

6.1. Specification

We proceed in two stages, each with its own specification. In the **static** stage we compare the two metrics on an equal footing. For each metric $\text{Metric} \in \{\text{LCRPGR}_{\log}, \text{BpCR}\}$ (we use the conservative log form of the ratio, Section 2) we estimate a two-way fixed-effects model without dynamics:

$$\text{Metric}_{i,t} = \beta_1 \ln(\text{UrbDens})_{i,t} + \beta_2 \text{UrbShare}_{i,t} + \beta_3 \ln(\text{GDPpc})_{i,t} + \beta_4 \text{NetMigr}_{i,t} + \mu_i + \delta_t + \varepsilon_{i,t}. \quad (1)$$

Because the data, controls, and fixed effects are held identical, any difference between the LCRPGR (log) and BpCR estimates of Equation 1 is attributable to the metric alone (H1).

In the **dynamic** stage, having established BpCR as the usable metric, we add its own lag to measure path-dependence, so the dependent variable is now specifically BpCR:

$$\text{BpCR}_{i,t} = \rho \text{BpCR}_{i,t-1} + \beta_1 \ln(\text{UrbDens})_{i,t} + \beta_2 \text{UrbShare}_{i,t} + \beta_3 \ln(\text{GDPpc})_{i,t} + \beta_4 \text{NetMigr}_{i,t} + \mu_i + \delta_t + \varepsilon_{i,t}. \quad (2)$$

The coefficient ρ measures how strongly a country’s per-capita land use persists from one period to the next (H5).

In both specifications the country fixed effects μ_i absorb all time-invariant national characteristics (geography, institutions, the initial level of urbanisation), so identification comes from changes *within* a country over time rather than from comparisons *between* countries. The period fixed effects δ_t absorb global shocks common to all countries in a given epoch (technology, the global business cycle, and any epoch-specific features of the satellite record). Standard errors are clustered by country to allow for serial correlation within a country’s series. The pooled models cluster on a large number of countries, so cluster-robust inference is reliable there; in the development-group splits of Section 7 the number of clusters is smaller (44 to 101 countries) and only eight periods are observed, so those subgroup standard errors, while still clustered, should be read as somewhat less precise.

Each regressor has a clear theoretical link to land consumed per resident. **Urban density** captures the compact-city mechanism: where settlement is already dense there is less room and less need to expand outward, so new residents are accommodated within the existing footprint (H2).² **GDP per capita** is ambiguous a priori: cross-country evidence links income to sprawl, while within a country income can fund densification; our design tests which force dominates (H3), a contrast developed in Section 8. **Net migration**, measured nationally, captures *international* migration: in-migrants settle in the existing stock, so demand is absorbed by densification faster than the footprint grows (H4); internal rural-to-urban migration is not observed (Section 8). Finally, the **lagged dependent variable** captures path-dependence, since the long-lived building stock, road network, and zoning make per-capita land use persist (H5). H1 concerns the metric and is addressed by comparing the LCRPGR and BpCR runs.

6.2. Dynamic-Panel Estimation and the Nickell Bias

Adding a lagged dependent variable to a fixed-effects model introduces a well-known problem. The within (demeaning) transformation that removes μ_i makes the transformed lagged regressor mechanically correlated

²The per-capita dependent variable and the per-area density control share the population term, so a common population shock could move both mechanically. The direction is informative: such a shock biases the density coefficient toward the negative sign we estimate (so H2 is read with caution) but biases the income coefficient toward positive, working *against* the negative H3 result, which is therefore conservative. As a check, entering density with a one-period lag (breaking the contemporaneous link) leaves income and migration unchanged but flips the density coefficient, confirming the compact-city estimate is partly mechanical whereas the income result is not. The built-up surface in the dependent variable is also only about a tenth of the urban land area in the control, so the two are far from collinear.

with the transformed error, biasing the estimate of ρ downward; the bias is of order $1/T$ and is therefore substantial in a panel with only eight periods such as ours. To correct it we use generalised method of moments (GMM) estimators designed for dynamic panels.

The **Arellano-Bond** (difference) estimator first-differences the equation to eliminate the country fixed effects and instruments the differenced lagged term with deeper lags in levels (lags two to five of BpCR). The **Blundell-Bond** (system) estimator augments this with a second, levels equation in which the lagged term is instrumented by its own past differences; this added moment information makes it more efficient when the series is persistent, as BpCR is. In both estimators only the lagged dependent variable is instrumented; the four controls are treated as strictly exogenous, which is a simplification, since income in particular is plausibly jointly determined with land use, so the control coefficients are best read as conditional associations rather than cleanly exogenous effects (Section 8). We treat **Arellano-Bond as the preferred specification** and report Blundell-Bond alongside it as a robustness comparison. Both are estimated in two steps with Windmeijer-corrected standard errors (Windmeijer, 2005), and the instrument matrix is collapsed (one column per lag distance) to keep the instrument count from proliferating with the time dimension.

The two estimators report observation counts that are not directly comparable to each other or to the 1,538 raw country-periods, and the difference between them is not arbitrary but follows exactly from how each equation is constructed. With collapsed GMM-style instruments, `plm:pgmm` substitutes zeros for instruments that are unavailable for a given country-period rather than dropping the row, a standard practice traced to Holtz-Eakin et al. (1988) and documented by Roodman (2009), under which the resulting instrument columns still correspond to a set of meaningful moment conditions rather than a distortion of the data. Consequently each country contributes a fixed number of rows to each equation, $T - k$, so that $N = 193 \times (T - k)$, where $T = 9$ is the number of epochs including the 1980 base period used only to form the 1985 growth rates (Section 4), and k is the number of periods the transformation costs: $k = 2$ for the differenced equation (one period lost because the lagged dependent variable itself is unavailable in the first epoch, one more lost to differencing) and $k = 1$ for the level equation (no differencing loss, only the lagged-dependent-variable period). This gives $N_{AB} = 193 \times 7 = 1,351$ and $N_{\text{level}} = 193 \times 8 = 1,544$, so that $N_{\text{level}} - N_{AB} = 193 \times (2 - 1) = 193$ exactly: the level equation recovers precisely one additional period per country, no more and no less. Exact counts are reported in Section 7.

Because the models are estimated in two steps, over-identifying restrictions are assessed with the robust Hansen J test, for which a p -value above 0.10 does not reject instrument validity. Serial correlation is checked with the Arellano-Bond **AR(1)** and **AR(2)** tests of the differenced residuals: by construction AR(1) should be significant and AR(2) should not, the latter being the condition under which the deeper lags are valid instruments. The collapsed instrument set is also kept well below the number of countries (193) to avoid the instrument-proliferation bias.

7. Results

7.1. The Metric Matters: LCRPGR vs. BpCR

We first compare the two metrics in the static (lag-free) two-way fixed-effects model of Equation 1, holding the data, controls, and fixed effects identical so any difference is due to the dependent variable alone. To keep specifications comparable, all models in this section (static and dynamic) use the common sample with the lagged dependent variable and all four controls available, so every column reports the same 1,499 observations and differences reflect the specification, not the sample. The contrast between the metrics is stark (Table 4).

With the official LCRPGR as the dependent variable, the model is essentially uninformative: the within R^2 is only 0.003 and not a single control is statistically significant, because the metric’s arithmetic tail outliers swamp any systematic variation. Replacing the dependent variable with BpCR, and changing nothing else, raises the within R^2 to 0.065, roughly 19 times higher, and the coefficients become significant and correctly signed: denser cities, richer countries, and higher in-migration are all associated with less land per capita. This is the clearest evidence for H1: the choice of metric, not the data, determines whether the relationships can be detected at all.

Table 4: Static two-way fixed-effects model on the estimation sample (1985-2020), same sample and controls for both metrics.

Variables	LCRPGR (log)	BpCR
ln(Urban Density)	-2.6079 (1.9467)	-0.0218*** (0.0070)
Net Migration (% Pop)	-0.0239 (0.0392)	-0.0008** (0.0003)
ln(GDP per capita)	+2.7401 (1.6817)	-0.0075*** (0.0028)
Urban Pop. Share	-14.9322 (16.1412)	-0.0014 (0.0261)
Observations	1,499	1,499
Within R ²	0.0033	0.0647
Country FE	Yes	Yes
Period FE	Yes	Yes

*** p<0.01, ** p<0.05, * p<0.1.

Standard errors clustered by country, in parentheses.

7.2. The Final Dynamic Model

Table 5: Final dynamic-panel model: FE stepwise build-up, Arellano-Bond (difference) and Blundell-Bond (system) GMM.

Variables	Dependent variable: BpCR						
	Fixed Effects					GMM (Bias-Corrected)	
	Base	+ Density	+ Net Migr.	+ GDP	+ Urban %	Arellano-Bond	Blundell-Bond
BpCR[t-1]	+0.2535*** (0.0578)	+0.2346*** (0.0582)	+0.2346*** (0.0585)	+0.2286*** (0.0583)	+0.2287*** (0.0583)	+0.6057*** (0.1081)	+0.5908*** (0.0623)
ln(Urban Density)		-0.0147** (0.0064)	-0.0146** (0.0065)	-0.0169** (0.0065)	-0.0165** (0.0065)	-0.0523*** (0.0142)	-0.0021** (0.0010)
Net Migration (% Pop)			-0.0008** (0.0003)	-0.0008** (0.0003)	-0.0008** (0.0003)	-0.0005* (0.0003)	-0.0010*** (0.0002)
ln(GDP per capita)				-0.0068** (0.0028)	-0.0068** (0.0028)	-0.0138** (0.0060)	-0.0004 (0.0003)
Urban Pop. Share					-0.0031 (0.0225)	-0.0310 (0.0520)	+0.0079* (0.0045)
Observations	1,499	1,499	1,499	1,499	1,499	1,351	1,544
Within R ²	0.076	0.091	0.108	0.125	0.125	–	–
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Period FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Instruments	–	–	–	–	–	15	21
Hansen (p)	–	–	–	–	–	0.44	0.14
AR(1) (p)	–	–	–	–	–	0.00	0.02
AR(2) (p)	–	–	–	–	–	0.71	0.47

Note: *** p<0.01, ** p<0.05, * p<0.1. Standard errors in parentheses: clustered by country in the fixed-effects columns; two-step GMM with the Windmeijer finite-sample correction (Windmeijer, 2005) in the Arellano-Bond and Blundell-Bond columns. The lagged dependent variable is instrumented by lags 2 to 5 of BpCR, with the instrument matrix collapsed. In the GMM columns the Observations row reports post-transformation equation observations (the differenced-equation set for Arellano-Bond; the larger level-equation set for Blundell-Bond) and is therefore not directly comparable across columns or to the 1,538 raw country-periods.

Recall that BpCR is an annual logarithmic growth rate of built-up area per capita, so a coefficient measures the change in that growth rate and the coefficients on logged regressors are semi-elasticities.

Path-dependence (H5). In the preferred Arellano-Bond estimate of Equation 2, the lagged term is large and highly significant, $\rho = 0.61$: about three-fifths of a country's per-capita land-use growth carries into the next period, and more than a third still remains two periods later ($0.61^2 \approx 0.37$). The GMM estimate is nearly three times the fixed-effects estimate ($\rho = 0.23$), exactly the direction and magnitude the Nickell bias predicts; sprawl, once underway, is sticky.

Urban density (H2). Log urban density enters at -0.052 and highly significant: holding income and migration fixed, a 10% higher density is associated with roughly 0.005 lower annual BpCR, sizeable against

a median BpCR of about 0.004: the compact-city mechanism, where dense settlement accommodates new residents by building up rather than out.

Income (H3). Log GDP per capita enters negative and significant at -0.014 : a doubling of income is associated with about 0.010 lower annual BpCR. This is the paper’s central substantive result: over time, as a country grows richer, its per-capita land use falls, consistent with income funding infill, vertical development, and tighter regulation (we contrast this with the cross-section in Section 8).

Net migration (H4). International net migration is negative and marginally significant (-0.0005): each additional percentage point is associated with slightly lower per-capita land use, consistent with in-migrants being absorbed into the existing stock faster than the footprint expands. This is the international channel only; internal migration is unobserved (Section 8).

Urban population share. The level of urbanisation is insignificant once the other drivers are controlled, indicating that what matters is the urban *form* (how densely a country builds) rather than the *stage* of urbanisation (how urban it already is).

The two estimators use 1,351 (Arellano-Bond) and 1,544 (Blundell-Bond) observations and 15 versus 21 instruments; the gap between the two counts is mechanical, not a contradiction (Section 6). Both pass the diagnostic tests: the Hansen J test does not reject the over-identifying restrictions ($p = 0.44$ and 0.14), AR(1) is significant while AR(2) is not (AR(2) $p = 0.71$ and 0.47), and the instrument counts (15 and 21) stay well below the 193 countries. Blundell-Bond confirms the headline results, near-identical persistence ($\rho = 0.59$) and a significant negative migration coefficient, but attenuates the density and income terms towards zero (income becomes insignificant); as its extra levels moments rest on a stationarity assumption a trending series can violate, we read the controls from the preferred Arellano-Bond column.

7.3. Heterogeneity by Development Group

Table 6: Heterogeneity: static two-way fixed-effects BpCR model by United Nations development group (same estimation sample as the static metric comparison above).

Variables	Dependent variable: BpCR		
	Developed	Developing	LDC
ln(Urban Density)	-0.0153* (0.0079)	-0.0182** (0.0086)	-0.0311** (0.0128)
Net Migration (% Pop)	-0.0020*** (0.0004)	-0.0010** (0.0004)	+0.0005 (0.0006)
ln(GDP per capita)	-0.0049** (0.0019)	-0.0070** (0.0030)	-0.0113* (0.0066)
Urban Pop. Share	-0.0348 (0.0274)	-0.0001 (0.0415)	-0.0036 (0.0355)
Observations	369	800	330
Within R ²	0.1883	0.0705	0.1019
Countries	48	101	44
Country FE	Yes	Yes	Yes
Period FE	Yes	Yes	Yes

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Standard errors are clustered at the country level, in parentheses, to robustify against potential serial correlation within subgroups.

The pooled model imposes a common slope on a very heterogeneous set of countries: the response of urban land use to income need not be the same in Germany as in Nigeria. Table 6 relaxes this by estimating the static model of Equation 1 separately for the three United Nations development groups, on the same estimation sample as Table 4. We use the static fixed-effects estimator rather than GMM here because the subgroups (44 to 101 countries) are too small for the deep-lag instruments to be reliable.

Three patterns stand out. First, the compact-city effect of density is present in every group but strengthens markedly from the developed (about -0.015 , significant only at the 10% level) through the developing (-0.018) to the least-developed countries (-0.031 , both at the 5% level): where cities are still forming, denser settlement absorbs new residents most strongly. Second, the income-densification result, the paper’s

central finding, carries its expected negative sign in all three groups, so it is not a rich-country artefact; it is estimated most precisely in the developed and developing groups (significant at the 5% level) and is larger but noisier in the small least-developed subsample (significant only at the 10% level). Third, net migration densifies in the developed and developing groups but is insignificant, and wrongly signed, for the least-developed countries, consistent with international migration mattering little there relative to the internal rural-to-urban flows that our national measure cannot capture.

7.4. Hypotheses

Taken together, the results broadly support the five hypotheses, with two qualifications noted below. The metric comparison supports **H1** (BpCR is the more reliable, better-fitting metric); the dynamic model supports **H3** (income reduces land per capita, within countries) and **H5** (per-capita land use is strongly path-dependent). **H2** (density reduces land per capita) is supported in sign and significance but should be read with caution: because the per-capita dependent variable and the per-area density control share the population term, part of the contemporaneous compact-city estimate is mechanical, and the coefficient flips when density is entered with a lag (see the footnote in Section 6). **H4** (international in-migration densifies) holds in the developed and developing groups but not in the least-developed countries, where the relevant internal rural-to-urban migration is unobserved.

8. Discussion

8.1. Interpretation and Relation to the Literature

The headline result is that, *within* a country, higher income is associated with *less* land consumed per urban resident. This only seems to contradict the established literature, in which richer countries are the more sprawled ones, because the two findings operate on different margins. *Across* countries at a point in time, higher income coincides with larger footprints through bigger dwellings and car-based mobility, the cross-sectional pattern documented by [Seto et al. \(2011\)](#) and [Oueslati et al. \(2015\)](#). *Within* a country over time, by contrast, growing richer is associated with densification, as income funds infill, vertical construction, redevelopment, and tighter land-use regulation. Our country fixed effects isolate this second margin, which cross-sectional designs cannot recover, so we extend rather than overturn the drivers literature: the sign of the income relationship depends on whether one compares countries or follows one over time. The development-group analysis in Table 6 sharpens this: the income relationship is negative in every group, but the compact-city density channel is markedly stronger in the developing and least-developed economies, where urban form is still being set, than in the already-dense developed group.

The results also bear directly on the measurement debate. [Nicolau et al. \(2019\)](#) argue on formal grounds that the official LCRPGR ratio is unstable, and [Lu and Weng \(2026\)](#) show that land-use-efficiency estimates are highly sensitive to the underlying data; our single-dataset metric comparison (Table 4) adds the complementary demonstration that the choice of metric, and not only the choice of data, can determine the conclusions one draws about sustainable urbanisation.

8.2. Limitations and Future Research

Several caveats apply. First, the estimates are associations, not causal effects: the GMM corrects the lagged-DV bias and the fixed effects absorb a large class of confounders, but the controls may still be correlated with omitted determinants of land use. Second, the five-year epochs capture medium-run trajectories rather than short-run adjustment. Third, the analysis area uses a fixed urbanisation threshold (GHS-SMOD of at least 21); alternative thresholds would shift the urban boundary and could affect the levels, if not the within-country patterns. Fourth, the satellite-based layers and national economic aggregates carry measurement error; in particular, net migration is national and so reflects international migration only, missing the internal rural-to-urban flows that often dominate urban expansion in developing countries, so the migration channel is likely understated. The panel also applies present-day boundaries retroactively: for the post-Soviet, ex-Yugoslav, and Czechoslovak successors, pre-dissolution quantities are reconstructed within today's borders, but a GDP control is unavailable wherever the country did not yet exist as a separate reporting unit. This leaves 37 country-periods across 29 countries without a GDP control, which complete-case estimation drops: 29 countries lack a 1985 value, namely 23 post-Soviet, ex-Yugoslav, and Czechoslovak successors that did not yet exist as separate reporting states, plus Eritrea, Ethiopia, Sudan, South Sudan, Timor-Leste, and Yemen, whose national accounts, on today's borders, similarly begin only after a later independence or border change; Sudan and South Sudan (which split in 2011) additionally lack

1990–2005 values. Finally, the pooled specification imposes a common slope, whereas Table 6 shows the income and density relationships, though consistent in sign, vary in strength across groups. These point to natural extensions: a sub-national analysis exploiting the ADM2 detail in the data, which would also measure internal migration directly;³ designs aimed at causal identification; and an assessment of BpCR as a practical, stable complement to LCRPGR in official SDG 11.3.1 monitoring.

9. Conclusion

Urban sprawl is real and measurable: across 1985–2020, built-up area per capita rose in almost two-thirds of country-periods. The difficulty is not the phenomenon but the yardstick. The official LCRPGR is unstable by construction, producing sign-flipping outliers exactly where land-use efficiency matters most; our log-based Built-up per Capita Rate is finite and interpretable for every country-period and reproduces the policy signal without the instability, raising within-country explanatory power by a factor of about 19 on identical data.

Using this stable metric on a dynamic panel of 193 United Nations member states, we find that per-capita urban land use is strongly path-dependent and, within countries, declines with urban density, net in-migration, and income, so that economic growth over time is accompanied by densification rather than sprawl. The income result is the more robust of the two associations; the density association, while consistent in sign, is partly mechanical (Section 6) and should be read as suggestive of the compact-city channel rather than as a clean causal estimate. The contribution is as much methodological as substantive. For monitoring sustainable urbanisation, the choice of metric is not a technicality but a precondition for drawing the right conclusions.

References

- Corbane, C., Politis, P., Siragusa, A., Kemper, T., Pesaresi, M., 2017. LUE User Guide: A tool to calculate the Land Use Efficiency and the SDG 11.3 indicator with the Global Human Settlement Layer. Technical Report JRC108026. Publications Office of the European Union. Luxembourg. DOI:10.2760/212689.
- FAO, 2024. The global administrative unit layers (GAUL) 2024. Food and Agriculture Organization of the United Nations. URL: <https://data.apps.fao.org/catalog/dataset/global-administrative-unit-layers-gaul-2024>. administrative boundaries, levels 0–2; provided under CC BY 4.0.
- Gao, J., O'Neill, B.C., 2020. Mapping global urban land for the 21st century with data-driven simulations and shared socioeconomic pathways. *Nature Communications* 11, 2302. DOI:10.1038/s41467-020-15788-7.
- Holtz-Eakin, D., Newey, W., Rosen, H.S., 1988. Estimating vector autoregressions with panel data. *Econometrica* 56, 1371–1395. DOI:10.2307/1913103.
- Lu, W., Weng, Q., 2026. From data inconsistency to reliable sdg 11.3.1 monitoring: A global multi-dataset assessment of land use efficiency. *Land Use Policy* 168, 108088. DOI:10.1016/j.landusepol.2026.108088.
- Nicolau, R., David, J., Caetano, M., Pereira, J.M.C., 2019. Ratio of land consumption rate to population growth rate: Analysis of different formulations applied to mainland Portugal. *ISPRS International Journal of Geo-Information* 8, 10. DOI:10.3390/ijgi8010010.
- Oueslati, W., Alvanides, S., Garrod, G., 2015. Determinants of urban sprawl in european cities. *Urban Studies* 52, 1594–1614. DOI:10.1177/0042098015577773.
- Pesaresi, M., Schiavina, M., Politis, P., Freire, S., Krasnodebska, K., Uhl, J.H., Kemper, T., et al., 2024. Advances on the Global Human Settlement Layer by joint assessment of Earth Observation and population survey data. *International Journal of Digital Earth* 17. DOI:10.1080/17538947.2024.2390454.
- Roodman, D., 2009. How to do xtabond2: An introduction to difference and system GMM in Stata. *The Stata Journal* 9, 86–136. DOI:10.1177/1536867X0900900106.
- Schiavina, M., Melchiorri, M., Pesaresi, M., 2023. GHS-SMOD R2023A - GHS settlement layers, application of the Degree of Urbanisation methodology (stage i) to GHS-POP R2023A and GHS-BUILT-S R2023A, multitemporal (1975–2030). URL: <http://data.europa.eu/89h/a0df7a6f-49de-46ea-9bde-563437a6e2ba>. DOI:10.2905/A0DF7A6F-49DE-46EA-9BDE-563437A6E2BA.
- Seto, K.C., Fragkias, M., Güneralp, B., Reilly, M.K., 2011. A meta-analysis of global urban land expansion. *PLOS ONE* 6, e23777. DOI:10.1371/journal.pone.0023777.
- Seto, K.C., Güneralp, B., Hutyra, L.R., 2012. Global forecasts of urban expansion to 2030 and direct impacts on biodiversity and carbon pools. *Proceedings of the National Academy of Sciences* 109, 16083–16088. DOI:10.1073/pnas.1211658109.
- UN DESA WPP, 2024. World population prospects 2024. <https://population.un.org/wpp/>.
- UN-Habitat, 2025. SDG Indicator 11.3.1 Metadata: Ratio of Land Consumption Rate to Population Growth Rate. SDG indicator metadata, harmonized template v1.1. United Nations Human Settlements Programme (UN-Habitat). URL: <https://unstats.un.org/sdgs/metadata/files/Metadata-11-03-01.pdf>. last updated 2025-04-23.
- UN M49 / UN-CDP, 2024. Standard country or area codes for statistical use (m49) and un-cdp least developed countries list. <https://unstats.un.org/unsd/methodology/m49/>.
- UN SNAAMA, 2025. National accounts main aggregates database: Gdp per capita, constant 2020 prices in us dollars. <https://unstats.un.org/unsd/snaama/>.
- Windmeijer, F., 2005. A finite sample correction for the variance of linear efficient two-step gmm estimators. *Journal of Econometrics* 126, 25–51. DOI:10.1016/j.jeconom.2004.02.005.
- Zhong, C., Peng, L., Yu, J., Swan, I., Li, H., 2025. Toward more reliable, complete, and equitable global urban land use efficiency assessments. *Communications Earth & Environment* 6, 1055. DOI:10.1038/s43247-025-03033-2.

³A first such case study, for Germany at the *Kreis* level, is provided as supplementary material at https://ofurkancoban.github.io/UrbanLandUseEfficiency_DevEconResearchProj/05_paper/supplementary.pdf.